1. Introduction

   This project focuses on predicting diabetes using the Diabetes Health Indicators Dataset. Three machine learning models were implemented: XGBoost, Random Forest, and K-Nearest Neighbors (KNN). The objective is to compare their performance and identify the best model for diabetes prediction while applying interpretability techniques.

2. Dataset Description

   The dataset used is the Diabetes Health Indicators Dataset from Kaggle. It contains health-related features such as BMI, HighBP, HighChol, Smoker, Stroke, Age, Income, and more. The target variable used is Diabetes_012 for multi-class diabetes classification.

3. EDA

   EDA included checking dataset shape, missing values, target distribution, correlation heatmap, BMI distribution, Age distribution, and relationships between diabetes and health indicators such as HighBP.

4. Data Preprocessing

   Preprocessing included removing duplicates, selecting the target variable, splitting the dataset into training and testing sets using an 80:20 ratio, and applying StandardScaler for normalization.

5. XGBoost Model

   Report XGBoost achieved the best performance among all models with high accuracy and strong interpretability using SHAP.

XGBoost Model

Model Overview

XGBoost (Extreme Gradient Boosting) is one of the most powerful supervised machine learning algorithms used for classification and regression tasks. It is based on boosting, where multiple weak learners (decision trees) are combined to create a strong predictive model.

In this project, XGBoost was used to predict diabetes using health indicators from the Diabetes Health Indicators Dataset.

It is known for high performance, fast execution, and excellent results in structured tabular datasets.

Why XGBoost Was Selected

XGBoost was selected because:

- It achieves very high prediction accuracy
- It handles missing values and noisy data efficiently
- It performs well on large datasets
- It reduces overfitting using regularization
- It works exceptionally well in medical prediction problems
- It supports explainability using SHAP

This makes XGBoost one of the best models for diabetes prediction.

Data Preparation for XGBoost

Before training the model, the following preprocessing steps were applied:

- Removed duplicate rows
- Selected Diabetes_012 as the target variable
- Split the dataset into training and testing sets using 80:20 ratio
- Applied StandardScaler for normalization

Although XGBoost does not strictly require scaling, normalization was applied to maintain consistency across all models.

Model Training

The model was trained using:

```
XGBClassifier(
    n_estimators=200,
    max_depth=6,
```

```
    learning_rate=0.05,
    random_state=42,
    eval_metric='mlogloss'
)
```

Where:

- n_estimators = 200 improves learning performance
- max_depth = 6 controls tree complexity
- learning_rate = 0.05 improves generalization
- eval_metric = mlogloss is suitable for multi-class classification

The model learned patterns from the training dataset and predicted diabetes classes on the testing dataset.

Model Evaluation

The XGBoost model was evaluated using:

- Accuracy Score
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC AUC Score

Results

Metric	Value
Accuracy	0.89
ROC AUC	0.84

Classification Report Summary

- Class 0 achieved excellent performance

- Class 1 remained difficult because of class imbalance
- Class 2 showed strong prediction improvement compared to KNN and Random Forest

XGBoost achieved the best overall performance among all models.

Confusion Matrix Analysis

The confusion matrix showed that:

- Most healthy patients were predicted correctly
- Prediction quality for minority classes improved compared to other models
- Fewer classification errors occurred overall

This demonstrates the strength of boosting methods in complex healthcare datasets.

SHAP Explainability (Interpretability)

SHAP (SHapley Additive exPlanations) was used to explain XGBoost predictions.

The most important features were:

- BMI
- HighBP
- Age
- GenHlth
- HighChol

SHAP summary plots showed how each feature positively or negatively influenced diabetes prediction.

This significantly improved model transparency and trust in medical applications.

Challenges Faced

Imbalanced Dataset

The main challenge was class imbalance.

Minority classes had significantly fewer samples, which affected recall and precision.

Possible Improvement

Applying SMOTE can improve class balance and further enhance minority class prediction.

Hyperparameter tuning can also improve model performance.

Strengths of XGBoost

- Highest prediction accuracy
- Excellent handling of complex patterns
- Strong resistance to overfitting
- Fast execution on large datasets
- Powerful explainability using SHAP

Limitations of XGBoost

- More complex than simpler models
- Requires parameter tuning
- Harder to explain without SHAP

Final Conclusion

XGBoost achieved the best performance in diabetes prediction among all implemented models.

It provided high accuracy, strong generalization, and excellent interpretability using SHAP.

Because of its performance and reliability, XGBoost is considered the best model for this project and the most suitable choice for real-world healthcare applications.

6. Random Forest Model Report
   Random Forest achieved strong performance and provided feature importance for interpretability

Random Forest Model

Model Overview

Random Forest is a supervised machine learning algorithm used for classification and regression tasks. It is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

In this project, Random Forest was used to predict diabetes based on health indicators from the Diabetes Health Indicators Dataset.

The model works by building many decision trees during training and selecting the final prediction based on majority voting.

Why Random Forest Was Selected

Random Forest was selected because:

- It performs very well on structured tabular datasets
- It handles non-linear relationships effectively
- It reduces overfitting compared to a single decision tree
- It provides feature importance for model interpretability
- It is highly reliable for medical classification problems

This makes it a strong candidate for diabetes prediction.

Data Preparation for Random Forest

Before training the model, the following preprocessing steps were applied:

- Removed duplicate rows
- Selected Diabetes_012 as the target variable
- Split the dataset into training and testing sets using 80:20 ratio

- Applied StandardScaler for feature normalization

Although Random Forest does not require scaling, normalization was applied for consistency across all models.

Model Training

The model was trained using:

```
RandomForestClassifier(
    n_estimators=200,
    max_depth=10,
    random_state=42
)
```

Where:

- n_estimators = 200 means 200 decision trees were created
- max_depth = 10 helps prevent overfitting
- random_state = 42 ensures reproducibility

The model learned patterns from the training dataset and predicted diabetes classes on the testing dataset.

Model Evaluation

The Random Forest model was evaluated using:

- Accuracy Score
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC AUC Score

Results

Metric	Value
Accuracy	0.86
ROC AUC	0.78

Classification Report Summary

- Class 0 showed excellent prediction performance
- Class 1 showed weaker performance due to class imbalance
- Class 2 showed moderate prediction quality

Random Forest performed better than KNN because it handles complex relationships more effectively.

Confusion Matrix Analysis

The confusion matrix showed that:

- Most non-diabetic cases were predicted correctly
- Minority diabetic classes were more difficult to classify
- Some confusion occurred between Class 1 and Class 2

This happened mainly because the dataset is imbalanced.

Feature Importance (Interpretability)

One major advantage of Random Forest is feature importance.

The most important features were:

- BMI
- Age
- HighBP
- GenHlth
- HighChol

This means these features had the strongest effect on diabetes prediction.

Feature importance improved model interpretability and helped explain how the model makes decisions.

Challenges Faced

Imbalanced Dataset

The main challenge was the imbalance between classes.

For example:

- Class 0 had a very large number of samples
- Class 1 had very few samples

This reduced model performance for minority classes.

Possible Improvement

Applying SMOTE can help balance the classes and improve recall for minority groups.

Strengths of Random Forest

- High prediction accuracy
- Handles complex relationships well
- Resistant to overfitting
- Provides feature importance
- Suitable for healthcare datasets

Limitations of Random Forest

- Slower than simpler models
- Harder to interpret than single decision trees
- Can still be affected by severe class imbalance

Final Conclusion

Random Forest achieved strong performance in diabetes prediction and performed better than KNN.

It successfully identified the most important health indicators affecting diabetes and provided strong classification results.

Due to its high accuracy and interpretability, Random Forest is considered one of the best models for this project.

7. KNN Model Report
   KNN provided acceptable performance but was affected by class imbalance and weaker minority class prediction.

KNN Model Report

K-Nearest Neighbors (KNN) Model

Model Overview

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification and regression tasks. In this project, KNN was used for diabetes prediction based on health indicators from the Diabetes Health Indicators Dataset.

KNN works by identifying the K closest data points (neighbors) to a new observation and assigning the most common class among them.

This model is simple, effective, and easy to interpret, especially for binary and multi-class classification problems.

Why KNN Was Selected

KNN was selected because:

- It is a simple and powerful baseline model
- It performs well on structured tabular datasets
- It does not require complex assumptions
- It helps compare performance with advanced models like XGBoost and Random Forest

Since the dataset contains normalized numerical features, KNN is a suitable choice.

Data Preparation for KNN

Before training the KNN model, the following preprocessing steps were applied:

- Removed duplicate rows
- Selected Diabetes_012 as the target variable
- Split data into training and testing sets using 80:20 ratio
- Applied StandardScaler for feature normalization

Feature scaling is very important for KNN because the algorithm depends on distance calculations.

Choosing the Best K Value

Different values of K were tested from 1 to 20 to determine the best number of neighbors.

A graph was plotted between:

- K Value
- Accuracy Score

The best performance was achieved using:

K = 5

This value was selected for the final KNN model.

Model Training

The model was trained using:

KNeighborsClassifier(n_neighbors=5)

The model learned patterns from the training dataset and then predicted diabetes classes on the testing dataset.

Model Evaluation

The KNN model was evaluated using:

- Accuracy Score
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC AUC Score

Results

Metric	Value
Accuracy	0.82
ROC AUC	0.65

Classification Report Summary

- Class 0 showed strong prediction performance
- Class 1 had weak performance due to very small number of samples
- Class 2 showed moderate performance

This happened because the dataset is imbalanced, where minority classes had significantly fewer samples.

Confusion Matrix Analysis

The confusion matrix showed that:

- Most healthy patients were classified correctly
- Minority diabetic classes were harder to predict
- Class imbalance affected recall and precision for rare classes

This is a common challenge in healthcare datasets.

Challenges Faced

Imbalanced Dataset

The main challenge was class imbalance.

For example:

- Class 0 had thousands of samples
- Class 1 had very few samples

This caused the warning:

UndefinedMetricWarning

because some minority classes were not predicted correctly.

Possible Solution

SMOTE (Synthetic Minority Oversampling Technique) can be applied to improve class balance and increase model performance.

Strengths of KNN

- Simple and easy to implement
- No training complexity
- Works well with normalized data
- Useful baseline model for comparison

Limitations of KNN

- Sensitive to imbalanced datasets
- Performance decreases with large datasets
- Requires feature scaling
- Slower prediction for very large datasets

Final Conclusion

The KNN model achieved acceptable performance in diabetes prediction, especially for the majority class.

However, due to class imbalance, performance on minority classes was limited.

Compared to XGBoost and Random Forest, KNN performed weaker, but it still provided a useful benchmark model for comparison.

Applying SMOTE and hyperparameter tuning could further improve the model in future work.

8. Explainability / Interpretability
   SHAP was used for XGBoost and Feature Importance for Random Forest to explain model decisions.
9. Model Comparison
   XGBoost performed best, followed by Random Forest, while KNN served as a useful baseline model.
10. Challenges
    The main challenge was class imbalance, which affected minority class prediction and evaluation metrics.
11. Conclusion
12. XGBoost was the best model for diabetes prediction and the most suitable choice for real-world healthcare applications